Mathematical Model of the Bistability in Embryonic Development

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2026 年 10 月 14 日

1 Mathematical Basis

Let's set the $x(t)$ = concentration of transcription factor x at time t.
So basically we can know that
$\boxed{\frac{dx}{dt} = \text{rate of synthesis} - \text{rate of loss}}$.
We can assume that every protein or transcription factor have same possibility of being degraded, so the rate of loss is
kx
where k is a constant with unit of s^{-1}.

1.1 When no Feedback from X itself

$\frac{dx}{dt} = a(S) - kx$
where $a(S)$ is the rate of synthesis at external signal S.
At the stable state, where $0 = \frac{dx}{dt} = a(S) - kx_0$.
$x_0 = \frac{a(S)}{k}$
Solve the differiential equation to obtain
$\int_{x(0)}^{x(t)} \frac{1}{a(S)-kx} dx = \int_0^t 1 dt$
$\Rightarrow x(t) = x_0 - (x_0 - x(0))e^{kt}$
which means that the $x(t)$ reaches the stable state **only** at x_0 when t is large enough. This result is independent from the initial transcription factor concentration $x(0)$.
So far, we obtained a monostability model. But what's different that contributes to a bistability model?

1.2 Positive Autoregulation Exists

The reason is simple, the X can affect the expression of X itself.
For example, the X activates its expression, increasing the rate of X synthesis.
Therefore, the rate of synthesis is now

$\frac{dx}{dt} = f(x, S) - kx$
and we assume the external signal is constant then the equation is now
$\frac{dx}{dt} = f(x) - kx$.
The form of $f(x)$ is unknown so far.

1.3 How rate of synthesis changes with X concentration

We can starts from the most simplified model, considering the binding of DNA and the transcription factor X.
$\mathrm{D} + \mathrm{X} \underset{k_{\mathrm{off}}}{\overset{k_{\mathrm{on}}}{\rightleftharpoons}} \mathrm{DX}$
$v_{\mathrm{on}} = v_{\mathrm{off}}$
$k_{\mathrm{on}}[D]x = k_{\mathrm{off}}[DX]$
Let $K_d = \frac{k_{\mathrm{off}}}{k_{\mathrm{on}}}$.
$[DX] = \frac{[D]x}{K_d}$
We will never know the concentration of binding points for X on the DNA. Since we need to convert $[DX]$ into a fractional occupancy $\theta = \frac{[DX]}{[D]+[DX]}$.
Consider the previous equations, we finally obtained: $\theta = \frac{x}{K_d+x}$.
This is the Hill function where Hill coefficient $n = 1$, showing a saturating relationship.
Then we assume the binding of X increase the rate of synthesis by r_d.
$r(x) = r_0 + r_d \frac{x}{K_d+x}$
When there are multiple X required for a single activation action. Similarly, we can finally obtain Hill equation with Hill coefficient n.
$H(x) = \frac{x^n}{K_d^n+x^n}$
When two X forms a dimer (or trimer, quatramer...) then bind to the DNA. The coefficient n will be 2 (or 3 or more respectively), but the K will depend on both affinity to DNA K_d and the affinity of forming dimer K_{dimer}.

1.4 Cooperativity

The binding mechanism of X can be very different. For example, the multiple binding site for X on DNA may have different affinity to X, or the binding of X affects the affinity for other binding sites. This **cooperativity** brings different Hill function.
Let's assume the binding of the first X increase the binding possibility by factor ω.
Let a relative weight of single binding of X $q = \frac{x}{K_d}$.
So the relative weight of double binding is ωq^2.
Let the weight of unbound DNA to be 1.
The possibility of double binding (we assume this activates the expression only) is
$P_{\mathrm{active}} = \frac{\omega q^2}{1+2q+\omega q^2}$.
Where ω showed the **cooperativity**.
$\omega = 1$ means two binding sites binds independently.
$\omega > 1$ means positive cooperativity.

$0 < \omega < 1$ means negative cooperativity.
This expression shows the probability of transcriptional activation under assumptions listed above. But the real gene regulatory system is definitely different. For example, a single binding may partially activate transcription. There are even more complex mechanisms behind this.
So generally, we can simplify this using a phenomenological Hill equation and allow the Hill coefficient n to be non-interger so that it can be a good approximation.
$H(x) = \frac{x^n}{K^n + x^n}$

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